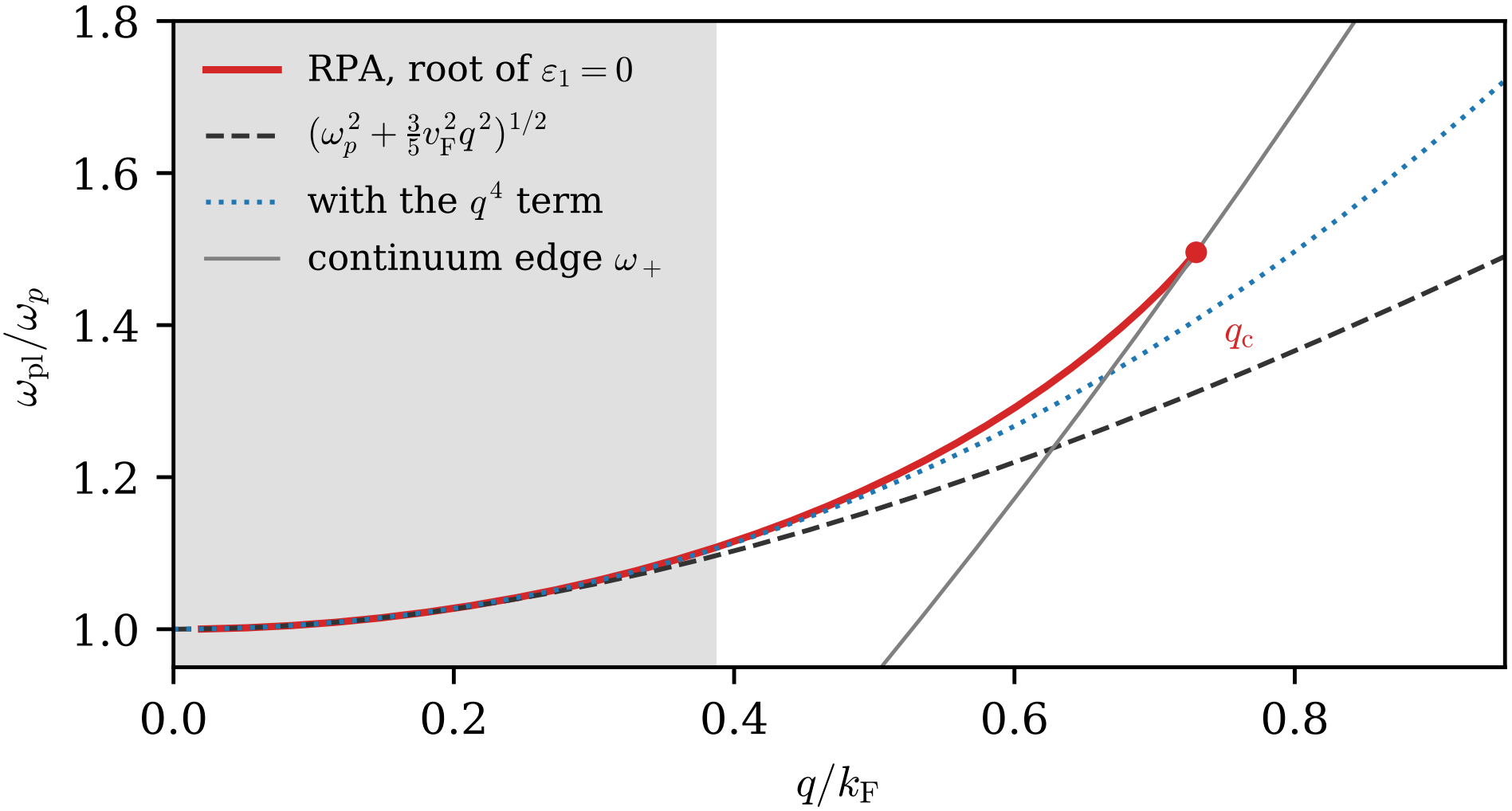
RPA, root of $\varepsilon_1 = 0$
$(\omega_p^2 + \frac{3}{5} v_{\mathrm{F}}^2 q^2)^{1/2}$
with the q^4 term
continuum edge ω_+
q_{c}
1.8
1.6
1.4
1.2
1.0
0.0
0.2
0.4
0.6
0.8
$\omega_{\mathrm{pl}}/\omega_p$
q/k_{F}